

Infinitely Many Aperiodic Greedy Pair-Sum-Avoiding Sequences

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Abstract

We construct an explicit infinite family of finite seeds whose greedy pair-sum-avoiding extensions have consecutive-gap sequences that are not eventually periodic. For every integer $p \geq 18$, the construction embeds the same scale-eight sumset controller in the residue classes $p, 2p, 4p$ modulo $7p$. A uniform finite shield gives a seed of cardinality $7p - 20$ and an infinite set S_p satisfying the exact greedy recurrence $t \in S_p \iff t \notin S_p + S_p$ beyond the seed.

1 Introduction and the main theorem

AI-use disclosure. This proof was developed with assistance from OpenAI's GPT-5.6 Sol. All arguments were checked and verified by the author.

The question appears in Erdős and Graham [1, p. 53] and is catalogued as Erdős Problem 341 in Bloom's online database [2]. Related periodicity criteria are due to Calkin and Finch [3], while Calkin and Erdős [4] proved that a natural class of aperiodic sum-free sets is incomplete. We answer the arbitrary finite-seed question negatively by constructing infinitely many counterexamples.

Write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. For $X, Y \subseteq \mathbb{N}_0$ and $c \in \mathbb{N}_0$, put

$$X + Y = \{x + y \mid x \in X, y \in Y\}, \quad c + X = \{c + x \mid x \in X\}, \quad X^c = \mathbb{N}_0 \setminus X.$$

Equal summands are allowed in every sumset.

Let $A = \{a_1 < \dots < a_k\} \subseteq \mathbb{N}$ be a nonempty finite set. Install all elements of A as the initial terms. For $n \geq k$, define

$$a_{n+1} = \min\{t \in \mathbb{N} \mid t > a_n, t \notin A_n + A_n\}, \quad A_n = \{a_1, \dots, a_n\}. \quad (1)$$

Let $d_m = a_{m+1} - a_m$ denote the consecutive gaps.

Fix an integer $p \geq 18$, put $M = 7p$, and work in $\mathbb{Z}/M\mathbb{Z}$, represented by $\{0, \dots, M-1\}$. Define

$$\begin{aligned} P &= \{p, 2p, 4p\}, & B &= \{M-3, M-1, 6p+3\}, & R &= P \cup B, \\ F &= R \cup (R-B) \cup (R-P), & D &= \mathbb{Z}/M\mathbb{Z} \setminus F, \end{aligned} \quad (2)$$

where $R-B = \{r-b : r \in R, b \in B\}$, and similarly for $R-P$. Set

$$A_p = D \cup B \cup (M+B) \cup \{p, 4p, M+2p, M+4p\}. \quad (3)$$

Theorem 1.1. *For every integer $p \geq 18$, the seed A_p in (3) has $7p - 20$ elements. For the sequence generated from A_p by (1),*

$$\forall K \geq 1 \ \forall h \geq 1 \ \exists m \geq K \quad d_{m+h} \neq d_m.$$

Thus its consecutive-gap sequence is not eventually periodic. In particular, the sets A_p , $p \geq 18$, form an infinite family of counterexamples.

Lemma 1.2 (Exact recurrence implies the greedy rule). *Let $S \subseteq \mathbb{N}$ be infinite, let $N \in S$, and put $A = S \cap [1, N]_{\mathbb{Z}}$. Suppose that*

$$t \in S \iff t \notin S + S \quad (t > N). \quad (4)$$

Then the increasing enumeration of S is exactly the greedy extension of A .

Proof. Scan the integers $t > N$ in order. Inductively, immediately before testing t , the selected set is $S \cap [1, t-1]_{\mathbb{Z}}$. Since all elements of S are positive, every representation $t = u + v$ with $u, v \in S$ has $u, v < t$, and hence

$$t \in (S \cap [1, t-1]_{\mathbb{Z}}) + (S \cap [1, t-1]_{\mathbb{Z}}) \iff t \in S + S.$$

Thus (4) makes the rule select exactly the elements of S , completing the induction. $\square$

2 A scale-eight sumset controller

For integers $a \leq b$, write $[a, b]_{\mathbb{Z}} = \{a, a+1, \dots, b\}$. We use the standard identity

$$[a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} = [a+c, b+d]_{\mathbb{Z}} \quad (a \leq b, c \leq d). \quad (5)$$

Define

$$L = \bigcup_{k \geq 0} [8^k, 4 \cdot 8^k - 1]_{\mathbb{Z}}, \quad (6)$$

$$Y_0 = \{0\} \cup \bigcup_{k \geq 0} [4 \cdot 8^k, 8 \cdot 8^k - 1]_{\mathbb{Z}}, \quad (7)$$

$$Y_1 = [1, 3]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [16 \cdot 8^k - 1, 32 \cdot 8^k - 1]_{\mathbb{Z}}, \quad (8)$$

$$Y_2 = [0, 1]_{\mathbb{Z}} \cup [7, 15]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [64 \cdot 8^k - 1, 128 \cdot 8^k - 1]_{\mathbb{Z}}. \quad (9)$$

At each scale, the corresponding blocks of L and $Y_0 \setminus \{0\}$ partition $[8^k, 8^{k+1} - 1]_{\mathbb{Z}}$. Thus

$$Y_0 = L^c. \quad (10)$$

Lemma 2.1 (Controller identities). *The sets in (7)–(9) satisfy*

$$Y_1 = (Y_0 + Y_0)^c, \quad (11)$$

$$Y_2 = (Y_1 + Y_1)^c, \quad (12)$$

$$Y_0 = \{0\} \cup (1 + (Y_2 + Y_2)^c). \quad (13)$$

Proof. A direct interval calculation gives

$$Y_0 + Y_0 = \{0\} \cup \bigcup_{k \geq 0} [4 \cdot 8^k, 16 \cdot 8^k - 2]_{\mathbb{Z}} = Y_1^c, \quad (14)$$

$$Y_1 + Y_1 = [2, 6]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [16 \cdot 8^k, 64 \cdot 8^k - 2]_{\mathbb{Z}} = Y_2^c, \quad (15)$$

$$Y_2 + Y_2 = [0, 2]_{\mathbb{Z}} \cup [7, 30]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [64 \cdot 8^k - 1, 256 \cdot 8^k - 2]_{\mathbb{Z}} = L - 1. \quad (16)$$

Here $L - 1 = \{\ell - 1 \mid \ell \in L\}$. Equations (16) and (10) yield (13). $\square$

Lemma 2.2. *The membership indicator of Y_0 is not ultimately periodic.*

Proof. Suppose that there are $N \geq 0$ and $h \geq 1$ such that

$$n \in Y_0 \iff n + h \in Y_0 \quad (n \geq N).$$

Choose $a = 8^k > \max\{N, h\}$ and put $m = 4a - h$. Then $N < a \leq m \leq 4a - 1$. Hence $m \in L$, so $m \notin Y_0$, whereas $m + h = 4a \in [4a, 8a - 1]_{\mathbb{Z}} \subseteq Y_0$. This contradicts the assumed periodicity. $\square$

3 A uniform finite shield modulo $7p$

All residues in this section lie in $\mathbb{Z}/M\mathbb{Z}$, with the notation of (2).

Lemma 3.1 (Uniform finite shield). *In $\mathbb{Z}/M\mathbb{Z}$, for every $p \geq 18$,*

$$D + B = \mathbb{Z}/M\mathbb{Z} \setminus R, \quad (17)$$

$$R \cap ((B + B) \cup (P + B) \cup (D + B) \cup (D + P)) = \emptyset, \quad (18)$$

$$(P + P) \cap R = P. \quad (19)$$

Moreover, the only ordered pairs in $P \times P$ whose sum lies in R are

$$p + p = 2p, \quad 2p + 2p = 4p, \quad 4p + 4p = M + p. \quad (20)$$

Proof. Expanding the definition in (2) gives

$$\begin{aligned} F = \{ & 0, 2, p - 6, p - 4, p, p + 1, p + 3, 2p - 3, 2p, 2p + 1, 2p + 3, \\ & 3p - 3, 3p - 1, 3p, 4p, 4p + 1, 4p + 3, \\ & 5p - 3, 5p - 1, 5p, 5p + 3, \\ & 6p - 3, 6p - 1, 6p, 6p + 3, 6p + 4, 6p + 6, M - 3, M - 2, M - 1 \}. \end{aligned} \quad (21)$$

For $p \geq 18$, these are 30 distinct representatives in $\{0, \dots, M - 1\}$. In particular, $|D| = M - 30$. The definition of F immediately gives

$$(D + B) \cap R = \emptyset, \quad (D + P) \cap R = \emptyset. \quad (22)$$

We prove the reverse inclusion in (17). Let $r \in \mathbb{Z}/M\mathbb{Z}$, and put $f = r + 1$. If $f \notin F$, then $f \in D$ and

$$r = f + (M - 1) \in D + B.$$

It remains to consider $f \in F$. Put

$$G = \{0, p + 1, 2p + 1, 4p + 1, 6p + 4, M - 2\}, \quad E = \{p - 6, 3p - 3, 5p - 3, 6p - 3, M - 3\}.$$

If $f \in G$, then direct subtraction shows that $r = f - 1 \in R$. If $f \in F \setminus (G \cup E)$, comparison with (21) shows that $f + 2 \notin F$, and hence

$$r = (f + 2) + (M - 3) \in D + B.$$

For the five values of f in E , the corresponding representations are

$$\begin{aligned} p - 7 &= (2p - 10) + (6p + 3), & 3p - 4 &= (4p - 7) + (6p + 3), \\ 5p - 4 &= (6p - 7) + (6p + 3), & 6p - 4 &= (M - 7) + (6p + 3), \\ M - 4 &= (p - 7) + (6p + 3). \end{aligned}$$

Thus every residue outside R belongs to $D + B$, and (22) proves (17).

Direct calculation shows that $B + B$ and $P + B$ are disjoint from R . Together with (22), this proves (18). Finally, the three mixed sums of elements of P are $3p, 5p, 6p$, all outside R , while the diagonal sums are those in (20). This proves (19) and the ordered-pair assertion. $\square$

We now define the infinite set

$$S_p = D \cup \{Mq + b \mid q \in \mathbb{N}_0, b \in B\} \cup \{Mq + p \mid q \in Y_0\} \cup \{Mq + 2p \mid q \in Y_1\} \cup \{Mq + 4p \mid q \in Y_2\}. \quad (23)$$

The initial controller memberships give

$$S_p \cap [1, 2M - 1]_{\mathbb{Z}} = D \cup B \cup (M + B) \cup \{p, 4p, M + 2p, M + 4p\} = A_p. \quad (24)$$

In particular, $2M - 1 = M + (M - 1) \in A_p$. The four sets in (24) are disjoint, and therefore

$$|A_p| = (M - 30) + 3 + 3 + 4 = M - 20 = 7p - 20. \quad (25)$$

Theorem 3.2. *For every $p \geq 18$ and every $t > 2M - 1$,*

$$t \in S_p \iff t \notin S_p + S_p. \quad (26)$$

Proof. Write $t = Mq + r$, with $0 \leq r < M$. Since $t > 2M - 1$, we have $q \geq 2$.

First let $r \in R = B \cup P$. The six possible pair-sum types are

$$D + D, \quad D + B, \quad D + P, \quad B + B, \quad B + P, \quad P + P.$$

A $D + D$ sum is at most $2M - 2 < t$. Equation (18) excludes the next four types from R , while (19) and (20) leave only the three diagonal $P + P$ pairs. Hence

$$\begin{aligned} Mq + 2p \in S_p + S_p &\iff q \in Y_0 + Y_0, \\ Mq + 4p \in S_p + S_p &\iff q \in Y_1 + Y_1, \\ Mq + p \in S_p + S_p &\iff q - 1 \in Y_2 + Y_2. \end{aligned}$$

Indeed, the corresponding representations are

$$\begin{aligned} (Mu + p) + (Mv + p) &= M(u + v) + 2p, \\ (Mu + 2p) + (Mv + 2p) &= M(u + v) + 4p, \\ (Mu + 4p) + (Mv + 4p) &= M(u + v + 1) + p. \end{aligned}$$

The exclusion of every other source type proves the converse implication in each equivalence. The controller identities (11)–(13) therefore make every P -rail integer represented exactly when it is absent from S_p . Every B -rail integer belongs to S_p , and the same exhaustive source analysis shows that none is represented. Thus (26) holds for $r \in R$.

Now let $r \notin R$. Such an integer t is not in S_p : its residue excludes every infinite rail, and $t > 2M - 1$ excludes the finite set D . By (17), there exist $d \in D$ and $b \in B$ such that

$$d + b = Mc + r \quad (c \in \{0, 1\}).$$

Since $q \geq 2$, we have $q - c \geq 1$, and

$$t = d + (M(q - c) + b) \in S_p + S_p.$$

The second summand is a positive element of a full B -rail. Both summands are positive and sum to t , so each is smaller than t . This completes the proof. $\square$

By (24), Theorem 3.2, and Lemma 1.2, the increasing enumeration of S_p is exactly the greedy extension of A_p .

4 Nonperiodicity of the gaps

Lemma 4.1. *Let $T = \{t_1 < t_2 < \dots\} \subseteq \mathbb{N}$ be infinite. If the consecutive gaps $t_{m+1} - t_m$ are eventually periodic, then the membership indicator of T is ultimately periodic.*

Proof. Suppose that the gaps have period $h \geq 1$ from index K onward. Let

$$H = \sum_{j=0}^{h-1} (t_{K+j+1} - t_{K+j}) > 0.$$

Every block of h consecutive gaps in the periodic tail has sum H , so $t_{m+h} = t_m + H$ for $m \geq K$. Since T is infinite, it is unbounded. Thus, given $n \geq t_K$, there is a unique $m \geq K$ such that $t_m \leq n < t_{m+1}$, and

$$t_{m+h} \leq n + H < t_{m+h+1}, \quad (n + H) - t_{m+h} = n - t_m.$$

Thus $n = t_m$ if and only if $n + H = t_{m+h}$, which proves

$$n \in T \iff n + H \in T \quad (n \geq t_K).$$

This is ultimate periodicity of the membership indicator. □

Proof of Theorem 1.1. Equation (25) proves the asserted cardinality. The residue- p rail of S_p satisfies

$$Mq + p \in S_p \iff q \in Y_0 \quad (q \in \mathbb{N}_0). \quad (27)$$

Suppose, for contradiction, that membership in S_p were ultimately periodic with period $h \geq 1$. Then Mh would also be an ultimate period. For all sufficiently large q , equation (27) would give

$$q \in Y_0 \iff Mq + p \in S_p \iff M(q + h) + p \in S_p \iff q + h \in Y_0,$$

contradicting Lemma 2.2. Thus the membership indicator of S_p is not ultimately periodic. The contrapositive of Lemma 4.1 now shows that the consecutive gaps in the increasing enumeration of S_p are not eventually periodic. By Lemma 1.2, this enumeration is the greedy extension of A_p , proving the theorem. □

References

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